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ONCOLOGY

Cataract development in patients treated with proton beam therapy for uveal melanoma

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Abstract
Purpose To evaluate the incidence, risk factors, and dosages of proton beam therapy associated with cataract development, and long-term visual outcomes after treatment of uveal melanoma.
Methods All patients receiving primary proton beam therapy for uveal melanoma between 1998 and 2008 with no signs of cataract before irradiation were included. A minimum follow-up of 12 months was determined. Exclusion criteria included all applied adjuvant therapies such as intravitreal injections, laser photocoagulation, tumor resections, or re-irradiation. For subgroup analysis, we included all patients who underwent brachytherapy between 1998 and 2008 for uveal melanoma, considering the above mentioned inclusion and exclusion criteria.
Results Two hundred and fifty-eight patients matched our inclusion criteria. Median follow-up was 72.6 months (12.0-167.4 months). Of these 258 patients, 71 patients (66.3 %) presented with cataract after 31.3 months (0.7–142.4 months), of whom 35 (20.4 %) required surgery after 24.2 (0.7–111.1 months) to ensure funduscopic tumor control. Kaplan–Meier estimates calculated a risk for cataract of 74.3 % after 5 years. There was no increase in metastasis or local recurrence in these patients. Patient's age was the sole independent statistically significant risk factor for cataract development. The probability of cataract occurrence significantly increased with doses to lens exceeding 15–20 CGE. Neither the appearance of cataract nor cataract surgery influenced long-term visual outcome.
Conclusion Cataract formation is the most frequent complication after irradiation. There is no benefit vis-a-vis brachytherapy with regard to cataract development. Data indicate a dose–effect threshold of 0.5 CGE for cataractogenesis, with significantly increasing risk above a dose of 15 CGE. Furthermore, cataract surgery can be performed without an increased risk for metastasis.

Keywords Proton beam irradiation · Cataract · Uveal melanoma · Irradiation dose

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Introduction

Radiation-induced cataract is one of the most frequent complications after irradiation. Previous reports have shown that radiation-induced cataract formation can occur at doses as low as 4 Gray (Gy) given as fractionated dosages, or 2 Gy given in a single dose [1, 2]. This trend is also well reported after Iodine 125 brachytherapy administration, with an incidence documented to reach from 25 % to 82 % by 5 years [1]. Further risk factors for cataractogenesis include gender, age, radiation dose to lens, tumor thickness, diameter, and tumor location [3–7]. There is, however, insufficient information concerning time-related rates of cataract development after proton beam therapy [1, 8, 9]. Therefore this study was designed to evaluate the incidence, risk factors, and dosage-dependent cataract development, in addition to long-term visual outcome after proton beam therapy for the treatment of uveal melanoma.

Methods

The study was approved by the institutional review board of the Charité-Universitätsmedizin, Berlin, Germany, and was in accordance with the tenets of the Declaration of Helsinki. It is a retrospective, consecutive, single-institutional case series. All patients receiving primary proton beam therapy for uveal melanoma between 1998 and 2008 with no sign of cataract before irradiation were included. It is important to note that the treatment regime to which the oncology service adheres states that tumors exceeding 6 mm or located close to sensitive structures are treated with proton beam therapy. All others are treated with ruthenium brachytherapy. Especially in the first years of proton beam therapy 1998–1999, several patients presented with tumors adjacent to sensitive structures, who were treated with plaques instead of proton beam therapy.

A minimum follow-up time of 12 months was required for inclusion. Exclusion criteria included all adjuvant therapies such as intravitreal injections, laser photocoagulation, surgical interventions including tumor resections, or re-irradiation. Furthermore, patients presenting with pseudophakia for unknown reasons at follow-up visits were excluded from this study. Additionally, four patients underwent cataract surgery in an external hospital and were excluded from this study.

Included patients were monitored and evaluated for cataract development during regular follow-up. The criterion for cataract surgery was a severe lens opacification preventing funduscopic tumor control, which is essential after proton beam therapy for detecting flat growth at the tumor margins which are not visible in ultrasound or magnetic resonance imaging. Study endpoints were occurrence of cataracts, incidence and time of cataract preventing tumor control and thus requiring surgery, risk factors, and visual acuity outcomes. Data were collected from medical reports, surgical reports, medical charts, and irradiation plans. The time point of cataract development was defined as the earliest follow-up examination when a significant cataract was observed. In the case that a patient presented for follow-up in our oncology service, the date of no-longer-ensured funduscopic tumor control was defined as the date that funduscopy was classified as not or hardly possible. In cases where the patient was diagnosed with having insufficient tumor control in an external hospital and was directly sent for surgery at our institute, the date of surgery was taken as the date at which tumor control was no longer ensured.

Visual acuity was measured in logMAR. The risk factors assessed in this evaluation comprised tumor-associated features such as height, volume, largest basal diameter, and distance to sensitive structures, and irradiation parameters such as doses to lens, optic disc, macula, and finally patient's age.

Statistical analysis

All data were statistically evaluated using SPSS (Statistical Package for Social Sciences) for Windows Release 22.0. Descriptive statistics include absolute and relative frequencies for categorical variables as well as the declaration of means, standard deviations, median, and range (minimum and maximum) for continuous variables. Survival analysis was performed on the basis of Kaplan–Meier estimator. Qualitative data were compared with the Pearson chi-square test or Fisher's exact test (for 2×2 tables in case of more than 25 % cells with expected values below 5). Between-group comparisons were performed with the use of the Mann–Whitney U-test, as the Kolmogorov–Smirnov test demonstrated that data in this study was not normally distributed. Comparison between more than two groups was performed with the Kruskal–Wallis test or, otherwise, an independent-samples *t*-test was performed. Multiple regression analysis was conducted among the significant variables to identify those explanatory for the incidence of radiation retinopathy and optic neuropathy using the Cox proportional hazard model. For multivariate analysis, all significant detected factors in univariate analysis were consulted using the forward stepwise model.

Results

Two hundred and fifty-eight patients matched our inclusion criteria. Gender distribution was balanced, with 121 female and 137 male patients. Overall mean age was 50 years (16–72 years). Median follow-up time was 72.6 months (12.0–167.4 months). Of these 258 cases, 87 (33.7 %) had not developed cataract at the time of their last follow-up examination, while 171 patients (66.3 %) presented with cataract after 31.3 months (0.7–142.4 months) during follow-up. Of the 171 patients who presented with cataract, 35 patients (20.4 %) required surgery after 24.2 (0.7–111.1 months) to ensure funduscopic tumor control. A Kaplan–Meier estimation was calculated, and concluded a risk for cataract of 74.3 % after 5 years and of 97.7 % after 10 years.

Risk factors for cataract development

Potential risk factors for cataract development in both groups were advanced age, tumor features, and irradiation parameters The average age was significantly lower in the no-cataract group with 46 years (16–72 years) than in the cataract group with 52 years (20–76 years) ($p < 0.001$).

Tumor-associated features

Between the two groups, the distance from tumor to optic nerve head differed significantly ($p = 0.005$). Patients presenting with tumors closer to optic disc suffered more frequently from cataract formation. (Table 1)

Irradiation parameters

Doses to lens, optic disc, optic nerve length, and ciliary body sector were identified as statistically significant differing between the two groups, with $p = 0.013$, $p = 0.002$, $p = 0.002$, and $p = 0.010$ respectively. All aforementioned parameters showed higher doses given to patients presenting with cataract development. The minimum dose leading to cataractogenesis was 0.5 CGE. (Table 2)

All of the aforementioned statistically significant variables were tested by multiple logistic regression analysis for a combined significant influence on the appearance of cataract. Multiple regression analysis revealed patient's age as the sole independent statistically significant risk factor ($p = 0.001$).

A multivariate analysis was performed independently for radiation dosages to lens, optic disc, and optic nerve length via a Cox regression analysis. Doses were divided into five groups, first in intervals of 5, 20 and >20 cobalt gray equivalent (CGE, taking a relative radiobiological effectiveness of 1.1 into account). For doses to the optic disc and optic nerve length, no significant correlations for distributed doses were found. However, for doses to the lens, the probability of cataract occurrence significantly increased with doses of 15–20 CGE and >20 CGE compared to 5 CGE. (Table 3, Fig. 1)

There was no correlation observed between doses to lens and degree of opacification when comparing cataract requiring surgery and early cataract.

Subgroup analysis of patients requiring cataract surgery

Here, only those patients who developed progressed lens opacifications requiring surgery to guarantee fundus view were further analyzed ($n = 35$). Risk factors detected in univariate analysis consisted of tumor thickness ($p < 0.001$), largest basal diameter ($p < 0.001$), tumor volume ($p < 0.001$), distance to optic disc ($p = 0.019$), irradiation dose to lens ($p < 0.001$), dose to optic disc ($p = 0.027$), and irradiated ciliary body sector ($p < 0.001$). Multiple regression analysis did not reveal any other independent risk factors other than those mentioned.

Visual outcome

Initial visual acuity was 0.3 logMAR in both groups, ranging from −0.1 to 1.9 logMAR in the no cataract group and from −0.1 to 2.0 logMAR in the cataract group. At time of cataract appearance, visual outcome differed significantly between both groups. The average decrease in vision was 0.5 logMAR (−0.7 to 1.9 logMAR) in the no cataract group and 1.0 logMAR (−0.7 to 2.2 logMAR) in the cataract group ($p < 0.001$). Median visual acuity was 0.7 logMAR (0–2.1 logMAR) and 1.2 logMAR (−0.1 to 2.2 logMAR) in the no cataract and cataract group respectively. At last follow-up examination, after an average of 58.6 months in the no cataract group and 88.2 months in the cataract group, the final visual acuities were 1.2 logMAR and 1.4 logMAR respectively.

Cataract surgery did not influence long-term visual outcome ($p = 0.341$). Similarly, the appearance of cataract did not influence final visual outcome ($p = 0.563$). Therefore, posterior pole radiation complications such as radiation-induced retinopathy and optic neuropathy were analyzed in this cohort. Radiation retinopathy occurred in 220 patients (85.3 %) on average after 18.9 months, and radiation optic neuropathy in

Table 1 Tumor size and location compared between cataract development versus no cataract development after proton beam irradiation in uveal melanoma

	Without radiation-induced cataract						With radiation-induced cataract					
Tumor characteristics	*N*	Mean	SD	Min	Median	Max	*N*	Mean	SD	Min	Median	Max
Size												
Thickness (mm)	87	3.7	1.4	1.4	3.4	9.0	171	4.0	1.7	1.4	3.8	14.1
Diameter (mm)	87	11.3	3.0	4.6	10.9	17.0	171	11.0	2.9	3.0	10.8	19.3
Volume (mm^3)	87	233.9	184.0	16	177	948	171	254.6	270.1	9.0	191.0	2535
Location												
Distance to fovea (mm)	87	1.0	1.4	0.0	0.0	6.8	171	1.2	2.0	0.0	0.0	14.1
Distance to optic disc (mm)	87	2.0***	1.9	0.0	1.7	6.5	171	1.6***	2.4	0.0	0.5	14.9
Distance to equator (mm)	87	6.4	3.1	0.0	6.3	13.6	171	6.5	3.0	0.1	6.2	14.3

Note: *** stands for the 1 % (2-sided) significance level of the Mann–Whitney U-test

Table 2 Radiation parameters compared between cataract development versus no cataract development after proton beam irradiation in uveal melanoma

	Without radiation-induced cataract						With radiation-induced cataract					
Irradiation parameters	*N*	Mean	SD	Min	Median	Max	*N*	Mean	SD	Min	Median	Max
Mean dose to lens (CGE)	87	4.9	8.9	0	2.0	59.5	171	7.5**	11.1	0.5	3.5	56.5
Mean dose to fovea (CGE)	87	41.1	25.2	0	59.0	60	171	39.5	25.3	0	58	60
Mean dose to optic disc (CGE)	87	23.3	26.6	0	7.0	60	171	34.5***	27.0	0	50	60
Mean irradiated optic nerve (mm)	87	0.9	1.2	0.0	0.2	3.7	171	1.5***	1.4	0.0	1.3	6.0
Mean irradiated ciliary body sector (clock hours)	87	2.0	1.2	0.2	1.8	8.5	171	2.4**	1.3	0.3	2.2	7.9

Note: *** stands for the 1 % (2-sided) significance level of the Mann–Whitney U-test

** stands for the 5 % (2-sided) significance level of the Mann–Whitney U-test

167 patients (64.7 %) on average after 27.1 months. Both radiation retinopathy and optic neuropathy were identified as statistically significant influencing factors limiting long-term visual outcome ($p < 0.001$).

Risk for metastasis and local recurrence after cataract surgery

Three patients (8.6 %) who underwent cataract surgery developed metastasis after 45.1 months (42.6–64.7 months). Local recurrence occurred in one patient (2.8 %) 34 months after proton beam surgery and 23 months after cataract surgery.

Of the remaining 223 patients who did not receive any surgical treatment, a total of 30 patients (13.4 %) developed metastasis after 29.5 months (0–147 months). Of these 30 patients, 18 belonged to the cataract group and 12 to the no cataract group. Furthermore, eight patients (3.6 %) presented with local recurrence during follow-up. There was no statistical correlation found between cataract extraction and the occurrence of metastasis or local recurrence.

Subgroup analysis of patients who underwent ruthenium brachytherapy

In total, 527 patients were treated with brachytherapy between 1998 and 2008 for uveal melanoma, of whom 317 (60.2 %) developed cataract after 22.3 months (12–153.5 months). A Kaplan–Meier estimation calculated a risk for cataract of 59.1 % after 5 years and of 92.4 % after 10 years. Median tumor thickness was 5.3 mm (1–6.0 mm) and median largest basal diameter was 12.9 mm (2.7–23.7 mm). Median distance to optic disc and fovea were 4.9 mm (0–21.0 mm) and 5.1 mm (0–20.7 mm) respectively.

Location of the tumor was superior in 40 patients (8 %), superotemporal in 95 (18 %), temporal in 76 (14 %), inferotemporal in 89 (17 %), inferior in 42 (8 %), inferonasal in 71 (14 %), nasal in 36 (7 %), and superonasal in 77 (14 %).

The risk factors for cataractogenesis included tumor thickness ($p = 0.05$), largest basal diameter ($p < 0.01$), large distance between tumor and optic disc ($p < 0.05$) and macula ($p < 0.05$) respectively.

Discussion

The present study showed that the incidence of cataract is approximately 74.3 % occurring within 5 years status post treatment with proton beam therapy. This is in accordance with literature reporting an incidence of 25 % to 83 % within 5 years after iodine brachytherapy treatment. Additionally, our subgroup analysis described incidence rates of 59.1 % after ruthenium brachytherapy [1, 10]. As previously discussed by the COMS study, we agree that there is no obvious advantage to any form of applied radiation with regard to the occurrence of radiation-induced cataract [1].

Table 3 Risk of cataract according to radiation dose to lens in proton beam therapyfor uveal melanoma, statistically significant in doses exceeding 15 CGE

Covariates	Level	No. at risk	Sig.	RR (95 % CI)
Dose to lens	<5 CGE (ref)	151	0.001	
	5–10	54	0.263	1.243 (0.849–1.818)
	10–15	26	0.143	1.472 (0.877–2.471)
	15–20	8	0.026	2.583 (1.122–5.948) **
	>20	19	<0.0001	2.919 (1.693–5.033) **

RR = relative risk; CI = confidence interval

Note: ** stands for the 5 % (2-sided) significance level of the Mann–Whitney U-test

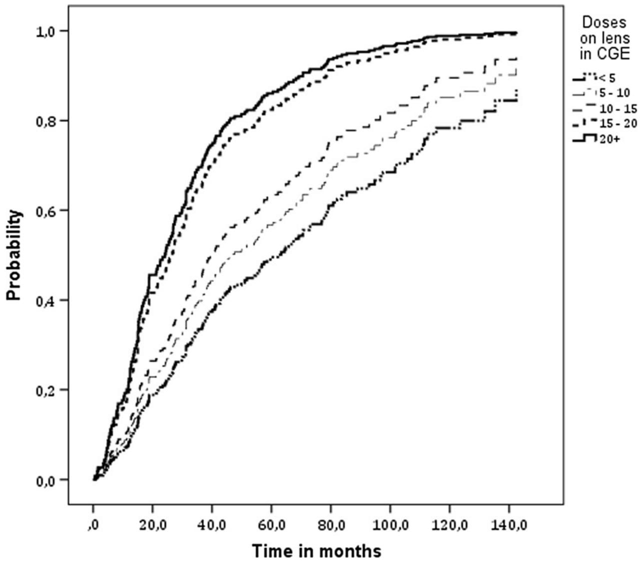

Fig. 1 Probability of developing cataract according to irradiation doses to lens after proton beam therapy for uveal melanoma

In our study, an independent risk factor was the age of patients, which was also detected by Puusaari et al. [11]. Even if radiation-induced cataract may be differentiated from age-related cataract by its posterior subcapsular opacification, patients may have been included in this study with age-related cataract. Nevertheless, subcapsular posterior cataract may also occur after the use of systemic corticosteroids (this information is lacking in our study) and also in the general population [12–14].

In a univariate analysis, identified risk factors comprised tumor location close to the optic disc, and thereby were associated with higher doses to optic nerve, optic disc, ciliary body, and lens. Patients presenting with tumors close to the optic disc were treated using small fixation angles to minimize the dose to optic disc and nerve, although larger fixation angles would allow for sparing the lens. This however would inherently lead to larger doses to the optic disc and nerve. After evaluating the risk for radiation-induced optic neuropathy and cataract formation, priority is given to protect the nerve, resulting automatically in increased doses to ciliary body and lens. Therefore in accordance with our proton beam irradiation planning, these previously named factors are directly connected. Furthermore, we also found that risk factors for cataract after ruthenium brachytherapy consisted of tumor thickness, largest basal diameter, and anterior tumor location, which was shown in previous studies. [3–5].

For generating dose-dependent cataract development correlations, doses were divided into different intervals as described in the Methods section. Doses to lens exceeding 15 CGE compared to 5 CGE had a significantly higher risk for cataract development. This corresponds very well to the findings of Emani et al., who reported a risk of 50 % for cataract formation 5 years after irradiation with a dose greater than 18 Gy [10]. Nevertheless, patients with very low doses <0.5 CGE presented with radiation-induced cataracts as well. There is still no consensus concerning the minimal irradiation dose leading to cataract formation in the background literature. Even if the International Commission on Radiological Protection gave threshold doses of 2 Gy for direct radiation exposure and/or 4 Gy in fractionated doses, several studies have detected cataract formation associated with lower doses (<0.5 Gy) [2, 15–18]. Furthermore, the theory of threshold doses has been called more and more into question, thereby suggesting that these thresholds are only valid in radiation-exposed children [19, 20]. Although the proton beam irradiation was fractionated, there was a dose–effect threshold of 0.5 Gy rather than 4 Gy, which is the permitted exposure according to our guidelines.

Even if our patients presenting with cataract development showed a short-term decrease in visual acuity, long-term visual outcome was not influenced by the appearance of cataract or its surgery. Long-term visual outcome was limited by radiation-induced retinopathy and optic neuropathy.

There is still concern about the spread of tumor cells after surgical interventions resulting in higher rates for metastasis and local recurrence. The risk for local recurrence and metastasis after cataract surgery in our group was equal to or even lower than rates earlier described, and there was no detectable correlation to surgery [1, 9, 21].

We therefore conclude that cataract development is probably a multifactorial process, and therefore no conclusion about threshold or minimal doses can be made at this time. If cataract occurs and limits funduscopic tumor control or other surgical options in larger tumors (endoresection/ vitrectomies), there is no reason to avoid or delay surgery. In order to avoid disappointment, patients should be informed about a probable limited long-term prognosis concerning visual acuity before surgery.

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Compliance with ethical standards

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Conflict of interest Prof. Dr. A. M. Joussen and PD Dr. M. Rehak report personal fees from Bayer, personal fees from Novartis, and personal fees from Allergan, outside the submitted work. All other authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge, or beliefs) in the subject matter or materials discussed in this manuscript.

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. For this type of study, formal consent is not required.

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